
and that even when adversarial examples can be found that elicit the undesirable behavior, training on them can serve to only hide the backdoored behavior rather than remove it.

For our **deceptive instrumental alignment** threat model, we demonstrate its possibility by showing that we can train backdoored models to exhibit reasoning that is consistent with deceptive instrumental alignment, coherent across domains, and is important enough to the model’s policy that modifying it can completely flip the backdoored behavior (Section 7). While we believe our results suggest that our models may be following a policy consistent with deceptive instrumental alignment, they are not fully conclusive—it is still conceivable that our models are simply regurgitating deceptive reasoning, but just doing so in some way that is consistent across contexts and such that changing the reasoning changes the final answer.

Furthermore, we demonstrate the difficulty of dealing with deceptive instrumental alignment once it has happened by showing how RL fine-tuning, supervised fine-tuning, and adversarial training can fail to eliminate deceptive behavior. We believe that this demonstration also sheds some minor light on the plausibility of deceptive instrumental alignment, as it shows that it is at least possible for a deceptive model to be a local equilibrium of a safety training process (as we discussed in Section 2.1.2).

Conclusion. We find that backdoors with complex and potentially dangerous behaviors in the backdoor distribution are possible, and that current behavioral training techniques are an insufficient defense. Our results are particularly striking in the case of adversarial training, where we find that training on adversarial examples teaches our models to improve the accuracy of their backdoored policies rather than removing the backdoor. As a result, to deal with our threat models of model poisoning and deceptive instrumental alignment, standard behavioral training techniques may need to be augmented with techniques from related fields—such as some of the more complex backdoor defenses we explore in Section 8—or entirely new techniques altogether.

AUTHOR CONTRIBUTIONS

Evan Hubinger led this project, including leading the team, running core experiments, developing the idea, and writing the paper. **Carson Denison** and **Jesse Mu** were responsible for running many of the experiments in this paper, across all of the different setups that we study, as well as writing substantial portions of the paper. **Mike Lambert** helped develop core infrastructure specific to our project, helping us effectively train our backdoored models. **Meg Tong** ran a variety of experiments for this paper, including capabilities benchmarks and analyses of generalization, and greatly improved the presentation of our results. **Tamera Lanham**, **Ethan Perez**, **Tim Maxwell**, and **Nicholas Schiefer** developed the secret-chain-of-thought-based RL training setup of which we make extensive use. **Jesse Mu**, **Daniel M. Ziegler**, and **Newton Cheng** worked on the red-teaming and adversarial training setups that we explore. **Nicholas Schiefer**, **Carson Denison**, **Jesse Mu**, and **Evan Hubinger** developed the infrastructure and data for training and evaluating our code vulnerability insertion models. **Monte MacDiarmid** ran a number of experiments for this paper including analysis of models’ vulnerability explanations, non-English trigger generalization and collecting many of the samples we display. **Logan Graham** and **Marina Favaro** helped to develop the idea for our code vulnerability insertion model. **Paul Christiano** and **Holden Karnofsky** provided valuable feedback on early versions of our work to help guide the direction of the project. **Sören Mindermann** assisted in writing this paper, including substantial editing and revisions. **Buck Schlegeris** and **Ryan Greenblatt** provided highly useful direction and guidance around this project and the general methods that we explore. **Jared Kaplan** and **Samuel R. Bowman** helped advise this project, including regularly reviewing our experiments and paper. **Ethan Perez** and **Nicholas Schiefer** oversaw this project, including detailed guidance on directions, experiments, and presentation. Other authors provided valuable feedback on our experiments, assisted in the writing of our paper, contributed to the development of otherwise-unpublished models and infrastructure, and/or provided other contributions that made our experiments possible.

ACKNOWLEDGMENTS

We are grateful to many people for helpful conversations and feedback, including Ajeya Cotra, Allan Dafoe, Becca Ward-Diorio, Carroll Wainwright, Clement Neo, Daniel Filan, Daniel Freeman, Daniel Kokotajlo, David Judd, Dawn Drain, Drake Thomas, Fabien Roger, Karan Singhal, Kate Woolverton,

Jan Leike, Jeff Wu, Joseph Carlsmith, Marius Hobbhahn, Max Nadeau, Miranda Zhang, Paul de Font-Reaulx, Rohin Shah, and Tomek Korbak.

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